Attention Sink in Transformers: A Survey on Utilization, Interpretation, and Mitigation

§2 Attention Sink in Transformers

- Preliminaries on Transformers (§2.1)
- Preliminaries on Attention Sink (§2.2)
- Language Models (§2.3)
  - Classical Language Models
  - Large Language Models
  - Mixture-of-Experts Large Language Models
  - Multi-Modal Large Language Models
- Vision Transformers (§2.4)
- Other Transformers (§2.5)

§3 Fundamental Utilization

- §3.1 Sink Token Preservation
- §3.2 Attention Redistribution
- §3.3 Learnable Prefix Tokens
- §3.4 Sink Token Repurposing

§4 Mechanistic Interpretation

- §4.1 Softmax Limitations and No-Op Theory
- §4.2 Outlier Circuits
- §4.3 Implicit Attention Bias
- §4.4 Geometric Anchoring
- §4.5 Other Mechanistic Interpretations

§5 Strategic Mitigation

- §5.1 Gated Attention Mechanisms
- §5.2 Modified Softmax Functions
- §5.3 Learnable Attention Bias
- §5.4 Pre-training Interventions
- §5.5 Other Mitigation Techniques

§6 Applications

- Model Pre-training
- Model Tuning
- Model Inference
- Mechanism Interpretability
- Reducing Hallucination
- Safety & Robustness
- General Capability Enhancement
- Long-Context Enhancement
- Multi-Modal Enhancement